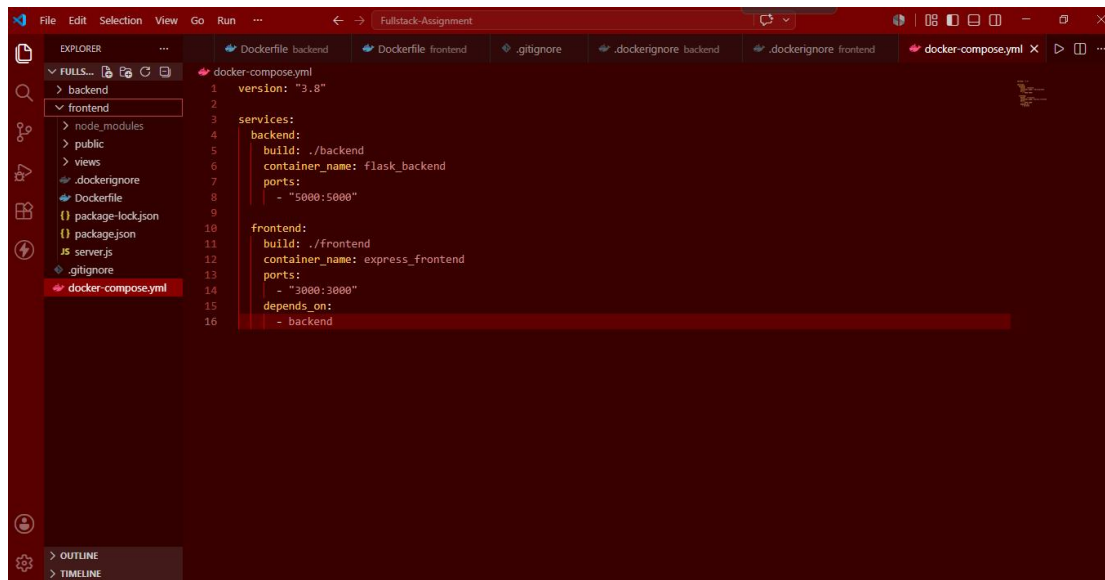


# Docker Assignment-5

## 1. Frontend (Node.js with Express):

- Create a frontend using Express and Node.js.
- Include a form similar to the one from the Flask Assignment 2.
- Configure the form to send a request to the Flask backend.
- **Backend (Flask):**
- Use the Flask backend to handle the form submission and process the data.
- **Folder Structure:**
- Organize the project with separate folders for the frontend and backend.
- **Docker Configuration:**
- Create a `Dockerfile` for both the frontend and backend.
- Write a `.yaml` file (Docker Compose) to connect both services in the same network.
- Upload both images to docker hub and push your whole code to github and add the `node_modules` and other non required files(`.vscode`) in `.gitignore`

## FrontEnd



The screenshot shows a VS Code editor window titled "Fullstack-Assignment". The Explorer sidebar on the left displays a project structure with folders for "backend", "frontend", "node\_modules", "public", "views", and files for ".dockerignore", "Dockerfile", "package-lock.json", "package.json", "server.js", ".gitignore", and "docker-compose.yml". The "docker-compose.yml" file is selected and its content is visible in the main editor area. The configuration defines two services: "backend" and "frontend".

```
1 version: "3.8"
2
3 services:
4   backend:
5     build: ../backend
6     container_name: flask_backend
7     ports:
8       - "5000:5000"
9
10  frontend:
11    build: ../frontend
12    container_name: express_frontend
13    ports:
14      - "3000:3000"
15    depends_on:
16      - backend
```

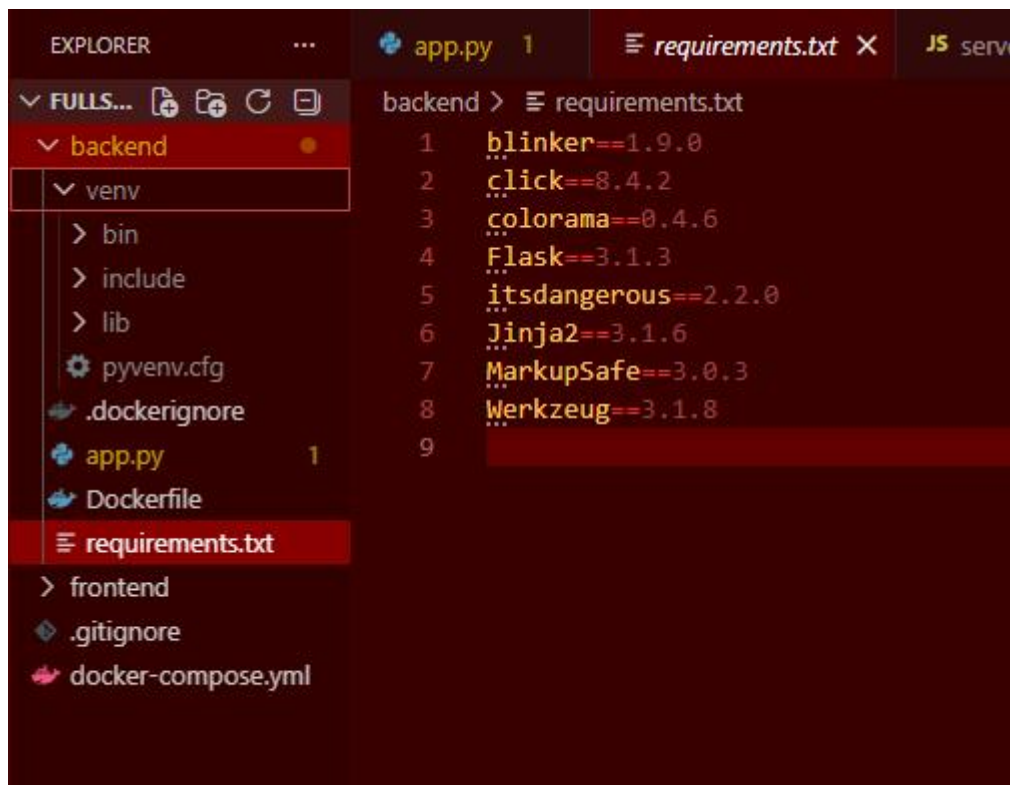
## Server.js

```
app.py JS server.js X index.ejs Dockerfile backend Dockerfile frontend .gitignore .dockerignore backend
frontend > JS server.js > app.post("/submit") callback
1  const express = require("express");
2  const bodyParser = require("body-parser");
3  const axios = require("axios");
4
5  const app = express();
6
7  app.set("view engine", "ejs");
8
9  app.use(bodyParser.urlencoded({ extended: true }));
10 app.use(bodyParser.json());
11
12 app.use(express.static("public"));
13
14 app.get("/", (req, res) => {
15   res.render("index");
16 });
17
18 app.post("/submit", async (req, res) => {
19
20   try {
21
22     const response = await axios.post(
23       "http://backend:5000/submit",
24       req.body,
25       {
26         headers: {
27           "Content-Type": "application/x-www-form-urlencoded"
28         }
29       }
30     );
31
32     res.send(response.data);
33
34   } catch (error) {
35     console.log(error.message);
36
37     if (error.response) {
38       console.log(error.response.data);
39     }
40
41     res.send("Error connecting to Flask Backend");
42   }
43
44 });
45
46 app.listen(3000, () => {
47   console.log("Frontend running on http://localhost:3000");
48 });
```

## Package.json

```
app.py JS server.js {} package.json X index.ejs Dockerfile backend
frontend > {} package.json > ...
1  {
2    "name": "Frontend",
3    "version": "1.0.0",
4    "main": "index.js",
5    "scripts": {
6      "test": "echo \"Error: no test specified\" && exit 1"
7    },
8    "keywords": [],
9    "author": "",
10   "license": "ISC",
11   "description": "",
12   "dependencies": {
13     "axios": "^1.18.1",
14     "body-parser": "^2.3.0",
15     "ejs": "^6.0.1",
16     "express": "^5.2.1"
17   }
18 }
19
```

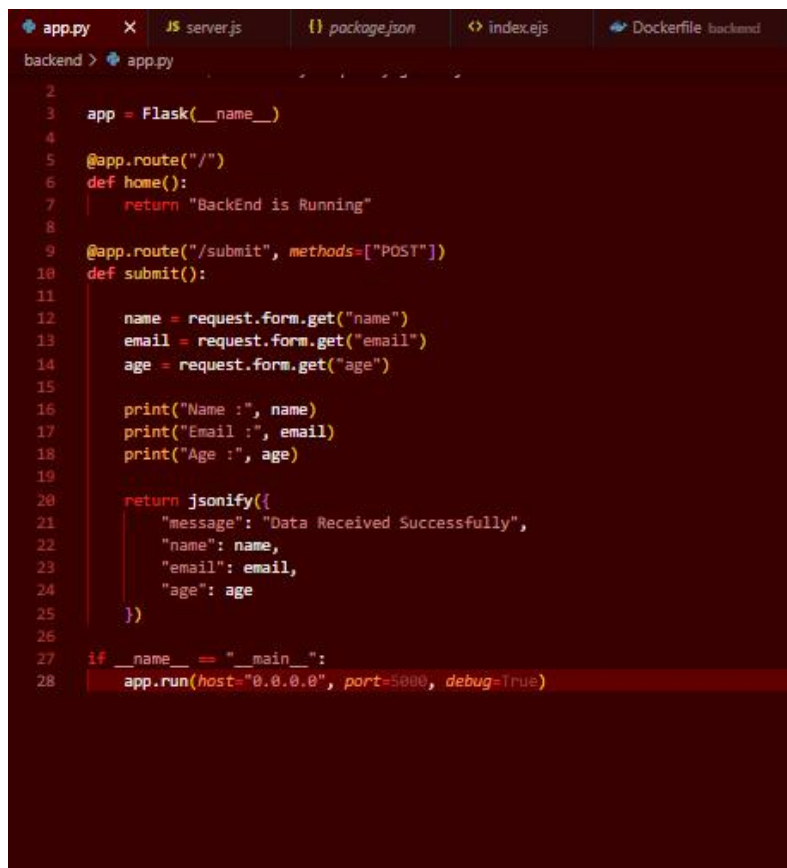
## BackEnd



The screenshot shows the VS Code interface. On the left, the Explorer sidebar displays a file tree with folders 'FULLS...', 'backend', and 'venv'. The 'backend' folder is expanded, showing files like 'bin', 'include', 'lib', 'pyvenv.cfg', '.dockerignore', 'app.py', 'Dockerfile', 'requirements.txt', 'frontend', '.gitignore', and 'docker-compose.yml'. The 'requirements.txt' file is selected and its contents are shown in the main editor area. The file lists dependencies for a Flask application.

```
backend > requirements.txt
1  blinker==1.9.0
2  click==8.4.2
3  colorama==0.4.6
4  Flask==3.1.3
5  itsdangerous==2.2.0
6  Jinja2==3.1.6
7  MarkupSafe==3.0.3
8  Werkzeug==3.1.8
```

## App.py



The screenshot shows the VS Code editor with the 'app.py' file open. The code defines a Flask application with two routes: a home page and a submit endpoint. The submit endpoint receives form data (name, email, age) and returns a JSON response. The application is configured to run on host '0.0.0.0' and port '5000'.

```
app.py
1
2
3  app = Flask(__name__)
4
5  @app.route("/")
6  def home():
7      return "BackEnd is Running"
8
9  @app.route("/submit", methods=["POST"])
10 def submit():
11
12     name = request.form.get("name")
13     email = request.form.get("email")
14     age = request.form.get("age")
15
16     print("Name :", name)
17     print("Email :", email)
18     print("Age :", age)
19
20     return jsonify({
21         "message": "Data Received Successfully",
22         "name": name,
23         "email": email,
24         "age": age
25     })
26
27 if __name__ == "__main__":
28     app.run(host="0.0.0.0", port=5000, debug=True)
```

## DockerFile Configuration-

### FrontEnd Dockerfile-

```
frontend > Dockerfile
1  FROM node:20
2
3  WORKDIR /app
4
5  COPY package*.json ./
6
7  RUN npm install
8
9  COPY . .
10
11 EXPOSE 3000
12
13 CMD ["node", "server.js"]
```

### Backend Dockerfile-

```
app.py 1 requirements.txt JS server.js <> index.ejs Dockerfile backend X
backend > Dockerfile
1  FROM python:3.11
2
3  WORKDIR /app
4
5  COPY . .
6
7  RUN pip install -r requirements.txt
8
9  EXPOSE 5000
10
11 CMD ["python", "app.py"]
```

## Docker-Compose.yml-

```
Dockerfile backend Dockerfile frontend .gitignore .dockerignore backend
docker-compose.yml
1  version: "3.8"
2
3  services:
4    backend:
5      build: ./backend
6      container_name: flask_backend
7      ports:
8        - "5000:5000"
9
10   frontend:
11     build: ./frontend
12     container_name: express_frontend
13     ports:
14       - "3000:3000"
15     depends_on:
16       - backend
```

## Docker Hub and Push

### Docker Compose

```
[+] Building 2/2
✓ fullstack-assignment-backend Built 0.0s
✓ fullstack-assignment-frontend Built 0.0s

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment
$ docker compose up
time="2026-07-02T02:34:17+05:30" level=warning msg="C:\\Users\\LENOVO\\Desktop\\Fullstack-Assignment\\docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] Running 3/3
✓ Network fullstack-assignment_default Created 0.1s
✓ Container flask_backend Created 0.6s
✓ Container express_frontend Created 1.4s
Attaching to express_frontend, flask_backend
flask_backend | * Serving Flask app 'app'
flask_backend | * Debug mode: on
flask_backend | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
flask_backend | * Running on all addresses (0.0.0.0)
flask_backend | * Running on http://127.0.0.1:5000
flask_backend | * Running on http://172.18.0.2:5000
flask_backend | Press CTRL+C to quit
flask_backend | * Restarting with stat
flask_backend | * Debugger is active!
flask_backend | * Debugger PIN: 105-984-628
express_frontend | Frontend running on http://localhost:3000
flask_backend | 172.18.0.3 - - [01/Jul/2026 21:07:41] "POST /submit HTTP/1.1" 200 -
[+] View in Docker Desktop View Config Enable Watch
```



## Docker Image

```
LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment
$ docker login

USING WEB-BASED LOGIN

Info → To sign in with credentials on the command line, use 'docker login -u <username>'

Your one-time device confirmation code is: TFZD-QFKZ
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser...

Login Succeeded

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment
$ docker images
```

| REPOSITORY                    | TAG                    | IMAGE ID     | CREATED        |
|-------------------------------|------------------------|--------------|----------------|
| fullstack-assignment-backend  | latest                 | 05d17cfc152d | 11 minutes ago |
| fullstack-assignment-frontend | latest                 | 956bd8cf58a9 | 12 minutes ago |
| python-hello-world            | latest                 | 9077e1a24ce7 | 13 days ago    |
| nginx                         | stable-alpine3.23-perl | a2528059d5b0 | 13 days ago    |
| nginx                         | latest                 | 6415da96b72a | 13 days ago    |
| alpine                        | latest                 | 28bd5fe8b56d | 2 weeks ago    |
| kalilinux/kali-rolling        | latest                 | 256893c92bbd | 2 weeks ago    |
| ubuntu                        | noble-20260509.1       | 786a8b558f7b | 6 weeks ago    |
| ubuntu                        | latest                 | f3d28607ddd7 | 2 months ago   |

```
LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment
```

## Docker Push and DockerHub

```
MINGW64:/c:/Users/LENOVO/Desktop/Fullstack-Assignment
LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop
$ cd Fullstack-Assignment

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment
$ docker push adityadev02/fullstack-assignment-frontend:v1
The push refers to repository [docker.io/adityadev02/fullstack-assignment-frontend]
7169c0aa9227: Layer already exists
a9f0592668cd: Layer already exists
d32999c9b84c: Layer already exists
6c4c846ffa95: Already exists
2f2ab6e4eb46: Layer already exists
0b22ec2d9b1a: Layer already exists
2933ecab0f11: Layer already exists
c83419f44309: Layer already exists
e66858429c4a: Layer already exists
76e6b8204a89: Layer already exists
db53381ee51f: Layer already exists
e5203b2bfeff: Layer already exists
8fcd2b4d799: Layer already exists
v1: digest: sha256:956bd8cf58a9203ada1d2588b4f281c976487b1be58e2ace9c8c7a6e758a4426 size: 856

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment
$ docker push adityadev02/fullstack-assignment-backend:v1
The push refers to repository [docker.io/adityadev02/fullstack-assignment-backend]
30d0db852850: Layer already exists
3f59c84a7863: Layer already exists
5543184d4729: Layer already exists
a71398e66418: Layer already exists
49364a62c77e: Layer already exists
40af14205d17: Already exists
aa3e9ef32f73: Layer already exists
0252e6abaf0f: Layer already exists
13f1d7323f0e: Layer already exists
f40567e9ead1: Layer already exists
06f788d60b9a: Layer already exists
v1: digest: sha256:05d17cfc152d6adec1097c66f376cd19024c4bf21a41d5275483857f64b851d6 size: 856

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment
$
```

## Repository

```
MINGW64:/c/Users/LENOVO/Desktop/Fullstack-Assignment
Initialized empty Git repository in C:/Users/LENOVO/Desktop/Fullstack-Assignment/.git/

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment (master)
$ git add .
warning: in the working copy of 'frontend/package-lock.json', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'frontend/package.json', LF will be replaced by CRLF the next time Git touches it

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment (master)
$ git commit -m "Full Stack Assignment Using Express , Flask and Docker"
[master (root-commit) cbeb2cc] Full Stack Assignment Using Express , Flask and Docker
12 files changed, 1213 insertions(+)
create mode 100644 .gitignore
create mode 100644 backend/.dockerignore
create mode 100644 backend/Dockerfile
create mode 100644 backend/app.py
create mode 100644 backend/requirements.txt
create mode 100644 docker-compose.yml
create mode 100644 frontend/.dockerignore
create mode 100644 frontend/Dockerfile
create mode 100644 frontend/package-lock.json
create mode 100644 frontend/package.json
create mode 100644 frontend/server.js
create mode 100644 frontend/views/index.ejs

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment (master)
$ git branch -M main

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment (main)
$ git remote add origin https://github.com/Aditya-Cyb/FullStack-Assignment.git

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment (main)
$ git push -u origin main
Enumerating objects: 17, done.
Counting objects: 100% (17/17), done.
Delta compression using up to 3 threads
Compressing objects: 100% (14/14), done.
Writing objects: 100% (17/17), 11.42 KiB | 324.00 KiB/s, done.
Total 17 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Aditya-Cyb/FullStack-Assignment.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop/Fullstack-Assignment (main)
$ |
```

# GitHub

